

An LLM-Based Simulation Framework for Benchmarking Reinforcement Learning in Health Interventions

Nolan Otam

Vrije Universiteit Amsterdam, Master of Science in Artificial Intelligence

Supervisor: Dr. Aneta Lisowska

Second assessor: Prof. dr. Mark Hoogendoorn

Abstract. Reinforcement learning (RL) is a natural framework for personalising adaptive health interventions, but the data needed to train and test such policies safely are rarely available, which motivates building a simulator. This thesis asks whether a large language model (LLM) can be the core of such a simulator, and how to decide whether the result is trustworthy for RL. Our framework represents synthetic patients as vignettes and personas, simulates their response trajectories with a zero-shot LLM coupled to warm-start anchoring and a two-part emission layer, and is evaluated through distributional realism, proximal action response, and policy transfer on a physical-activity trial, an adherence trial, and a controlled synthetic benchmark. The results show that a policy trained entirely inside the simulator transfers to the known-effect reference environment, recovering 94% of the return of the same learner trained directly on that environment. The simulator also ranks candidate policies only coarsely: across 49 policies the correlation with true return is moderate ($\rho=0.66$) but concentrated among the poor policies and collapses to near zero among the best, a selection ability gated by the environment's partial observability. Finally, useful policy training does not require indistinguishable trajectories: the simulated data remain distinguishable from real data, yet still suffice to train the transferable policy. One fidelity limit remains beyond this: the model over-states the action response, producing a positive proximal contrast where the observed contrast is effectively zero. This limitation matters most in health, where genuine effects are small. In this controlled testbed, the LLM simulator is useful for policy training and coarse policy screening. Its trustworthiness therefore depends on decision-level validation in addition to trajectory realism.

Keywords: Reinforcement learning · Large language models · Synthetic patient simulation · Mobile health · Just-in-time adaptive interventions · Simulator validation.

1 Introduction

An important class of artificial-intelligence applications in healthcare must adapt to a single person over time. Mobile-health applications repeatedly decide whether to prompt a user with a reminder or a suggestion, and the usefulness of each decision depends on the person’s recent behaviour, context, and response to earlier prompts. Such just-in-time adaptive interventions are inherently sequential. An action taken now influences not only the immediate outcome but also the user’s future receptiveness, and over-prompting erodes engagement over time [22,15]. This makes reinforcement learning (RL), in which an agent observes a state, chooses an action, and adjusts its behaviour from feedback, the natural computational framework.

The problem is consequential: roughly half of patients with chronic illness do not follow treatment recommendations, with substantial consequences for health outcomes and healthcare costs [30]. Yet applying reinforcement learning here is constrained by a shortage of data. Behavioural-intervention studies typically enrol a few dozen participants (the REINFORCE adherence trial that motivates this work enrolled about sixty despite substantial infrastructure [16]); trials are expensive and slow, and the resulting clinical data are subject to privacy constraints that make sharing them difficult. Free exploration on real patients is, moreover, neither practical nor ethical: a poorly behaved policy can contact users at the wrong moments and erode their trust. A common response is to build a simulator: a model of how patients respond to interventions, inside which candidate policies can be developed and compared without exposing patients to exploratory actions.

This raises the question of how to build such a simulator when very little real data are available. Large language models are an appealing candidate because they carry broad prior knowledge about human behaviour that a small dataset cannot supply [4,10], and recent work uses them to generate synthetic patients and simulate behaviour [23,1,26]. In this thesis, those priors are accessed through prompting rather than task-specific fitting, avoiding dependence on the trajectories that are missing in the first place. This introduces the separate difficulty of establishing whether the resulting simulator is trustworthy. Trajectories that look realistic are not necessarily safe to optimise a policy against [20,21], and establishing how to measure that trustworthiness is a thread that runs through the thesis.

The project was originally framed around the REINFORCE medication-adherence trial [16], whose day-level records proved unsuitable as a reference distribution (Section 3.7); the empirical work was therefore reframed around the public HeartSteps physical-activity trial. This remains a demanding testbed because the proximal effect of an activity suggestion is small and often statistically indistinguishable from zero (Section 4). Adherence remains a motivating example, while the framework itself is domain-agnostic. The central research question is: can an LLM-based synthetic patient simulator provide a reliable benchmarking environment for reinforcement-learning policies targeting personalised adaptive health interventions? We approach it through three specific research

questions. **RQ1:** can an LLM, coupled with a suitable emission model, produce response trajectories that statistically resemble real ones? **RQ2:** beyond looking realistic, does the simulator capture the decision-relevant structure of the data, including the differences between patients and the proximal response to the intervention? **RQ3:** how does a policy trained inside the simulator perform when evaluated against an external reference target, and can the simulator’s ranking of candidate policies be trusted?

Contributions.

- **An LLM simulation environment:** structured patient profiles, zero-shot trajectory generation, warm-start anchoring, and a two-part (hurdle) emission layer, assembled into one benchmarking environment (Section 3, Figure 1).
- **A decision-focused validation methodology:** distributional realism, proximal action response, and policy transfer are used to judge the simulator by the decisions it would induce; applied to a public micro-randomised trial, it reveals a persistent discrepancy: the simulator over-states the observed action response (Sections 3.6, 4.2).
- **A train-on-synthetic, test-on-reference study:** a policy trained entirely inside the simulator recovers 94% of the return of the same learner trained on the known-effect reference target, whereas its ranking of 49 policies resolves the poor policies far better than the best ones, an ability gated by the target’s inference noise (Section 4.4).

The originality claimed is accordingly not the use of an LLM to simulate patients, which is already the subject of a growing literature (Section 2), but the validation of an LLM as an environment by the decisions it induces: distributional realism, action-response fidelity, policy transfer, and ranking fidelity.

2 Related Work

Reinforcement learning for adaptive health interventions. Just-in-time adaptive interventions formalise mobile health as a sequence of decision points at which a policy chooses whether and how to prompt a user [22]. Micro-randomised trials randomise the action at each decision point so that proximal causal effects can be estimated [14], with HeartSteps the canonical physical-activity example [15]; reinforcement learning has been deployed on it directly, the central difficulty being to learn a personalised policy from few users and weak, noisy proximal effects [18], and the REINFORCE trial illustrates the same constraint on the adherence side [16]. This scarcity motivates developing and evaluating policies inside a simulator rather than on people; offline reinforcement learning studies learning from a fixed, possibly biased batch of logged experience [17], and off-policy evaluation scores a policy from such logged data without deploying it [25,28].

Language models as behavioural simulators. Language models carry enough prior knowledge of human behaviour to substitute for human respondents in some tasks: they reproduce human choice patterns in classical cognitive tasks [4], act as simulated economic agents [10], emulate the response distributions of human subgroups with measurable “algorithmic fidelity” [2], replicate canonical human-subject experiments [1], produce believable individual and emergent social behaviour in a multi-day agent sandbox [23], and recover an individual’s survey responses from their own interview [24]. In health, LLMs have been conditioned on structured patient vignettes and profiles to drive simulated patients [26,31], while other work uses them to generate medical case vignettes [3]. This literature establishes persona-conditioned behavioural simulation as feasible; what it does not address, and this thesis examines, is whether the resulting trajectories are calibrated enough to train a policy against.

Simulators, trust, and positioning. Training a policy inside a learned generative model of an environment and transferring it outward is the world-models idea [8]; the matching evaluation principle, train on synthetic and test on real, originates in medical time-series generation [6]; and language models have been cast as world models for planning [9], the framing closest to ours, though there the model plans over its own predictions whereas we use it as the environment for a separately trained policy. To quantify how distinguishable simulated trajectories are we use the classifier two-sample test [19]; in our implementation it operates on selected trajectory-summary features, so near-chance accuracy supports similarity only with respect to those features and the classifier used. A cautionary line of work warns that believable behaviour is not accurate behaviour and must be checked with objective, action-level metrics [20]. Closest in spirit, Martinson et al. complement predictive evaluation with uncertainty quantification and decision-focused analysis [21]; we adopt that stance and extend it to a transfer test. StepCountJITAI [11] serves as a comparator and controlled testbed with a known effect, but is methodologically the opposite of our approach, with hand-specified, equation-based dynamics and prespecified parameter distributions, so we contrast against it rather than build on it. Our contribution, to our knowledge novel in combination, is to put the language model in the environment through structured vignette-and-persona conditioning, zero-shot LLM-driven trajectory simulation, and a principled two-part (hurdle) emission layer [5], starred in Figure 1. The primary contribution is methodological: a validation protocol pairing distributional-realism checks [19] with action-response and controlled reference-transfer tests, adapting the train-on-synthetic/test-on-real tradition [6,8].

3 Methodology and Experimental Setup

The framework consists of the generation of synthetic patients as vignettes and personas, the simulation of their response trajectories with an LLM and an emission layer, and the benchmarking of policies inside the resulting environment (Figure 1). This section describes each part, the evaluation methodology, the datasets, and the experimental design.

3.1 Problem formalization

We model a just-in-time adaptive intervention as a finite-horizon sequential decision process, following the standard formulation of reinforcement learning [27] and of micro-randomised mobile-health trials [22,14]. For a given patient, decisions are taken at points $t \in \{1, \dots, T\}$. At each point the controller observes a *state* $s_t \in \mathcal{S}$, selects an *action* $a_t \in \mathcal{A}$, the patient returns a scalar *outcome* $y_t \in [0, 1]$, and the controller receives a task-specific reward r_t . In the LLM world $r_t = y_t$; in the RQ3 transfer test, the external reference supplies its native step-count reward. The components are:

- **State** $s_t \in \mathcal{S}$: the information exposed to the policy. In RQ3 the policy state consists of the current observable context label (inferred during reference evaluation) and the number of consecutive nonzero messages, capped at three. Trajectory history is used separately by the LLM simulator when generating the next outcome (Section 3.7).
- **Action** $a_t \in \mathcal{A}$: the finite intervention set; for HeartSteps `{no_suggestion, active_suggestion, sedentary_break_suggestion}`, for StepCountJITAI the four-message set of Section 3.7.
- **Outcome** $y_t \in [0, 1]$ **and reward** r_t : the simulator emits a normalised proximal response (a scaled thirty-minute step count for HeartSteps or a weekly coverage value for HPTN067) and uses it as its training reward; StepCountJITAI supplies the resulting step count as its reference reward.

A stationary policy $\pi_\theta : \mathcal{S} \rightarrow \mathcal{A}$, with parameters θ , is evaluated by its expected return $J(\pi_\theta) = \mathbb{E}_{\pi_\theta} [\sum_{t=1}^T r_t]$, the quantity a reinforcement-learning agent maximises.

The framework supplies a generative model of the environment: a simulator that produces the outcome y_t in place of a real patient, so that policies can be trained and compared without contacting people. A patient is represented by a profile $P = (V, \rho)$ pairing a structured vignette V with a free-text persona ρ (Section 3.2). At decision point t the language model maps the available conditioning to a latent expected tendency

$$\ell_t = \text{LLM}(c_t, a_t, P, A_{1:K}, H_{K+1:t-1}) \in [0, 1], \quad (1)$$

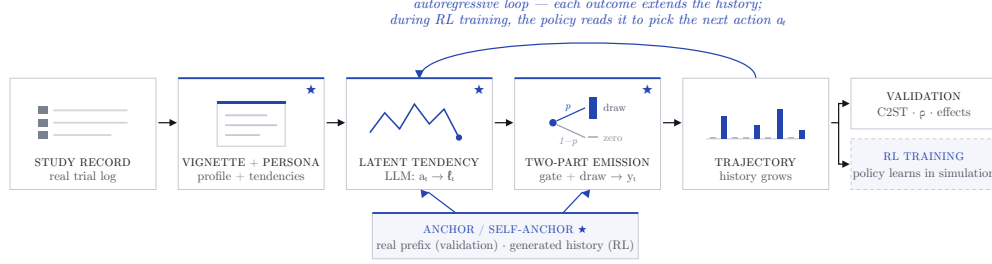
where c_t is the current context; $a_t \in \mathcal{A}$ the action under evaluation; P the patient profile; $A_{1:K} = (y_1, \dots, y_K)$ the warm-start anchor, the patient’s first K real outcomes used only as conditioning (Section 3.4); and $H_{K+1:t-1} = (y_{K+1}, \dots, y_{t-1})$ the outcomes the simulator has itself produced so far. The latent ℓ_t is the model’s smooth estimate of the expected response and is not emitted directly: a two-part emission operator $\mathcal{E}$ converts it into the realised outcome

$$y_t \sim \mathcal{E}(\ell_t; r_b, \bar{b}), \quad (2)$$

where $r_b \in [0, 1]$ and $\bar{b} \in [0, 1]$ are the positive-response rate and mean of the anchor outcomes $A_{1:K}$ (defined in full in Section 3.5). Applying Equations (1) and (2) autoregressively, appending each sampled y_t to the history H , defines the

Table 1. The pipeline’s input–output contract, one row per stage of Figure 1; t indexes decision points and all symbols are defined in this section.

Stage	Inputs	Operation	Output	Role
Patient profile	study records / sampled traits	vignette + persona construction	$P = (V, \rho)$	conditioning
LLM latent model P, c_t, a_t , optional validation anchor $A_{1:K}$, generated history		zero-shot prediction, Eq. (1)	$\ell_t \in [0, 1]$	behavioural prior
Emission layer	$\ell_t, r_b, \bar{b}$	hurdle/log-normal draw, Eqs. (4)–(6)	outcome y_t	realised trajectory
Evaluation / RL	simulated, real and reference trajectories	C2ST, effect, transfer, ranking	validation verdict	benchmark trust

**Fig. 1.** The simulator pipeline. Starred stages are the novel components; at each decision point the model predicts a latent tendency ℓ_t from vignette, context and simulated-so-far history, the emission layer turns it into a realised outcome, and the trajectory builds autoregressively (top loop, Equations (1)–(2)). For real-data validation, the first K observed outcomes anchor the model and emission; synthetic RL populations instead self-anchor on generated outcomes. The final stage splits into validation (C2ST, effects, transfer) and RL training, where the policy reads its observable state to pick the next action.

simulated transition dynamics, an LLM-parameterised environment in which a policy is trained and then frozen. Writing $\hat{J}(\pi)$ for the return measured inside the simulator and $J(\pi)$ for the true return on the target of interest, the validation question (Section 3.6) is whether $\hat{J}$ is a trustworthy proxy for J , particularly whether it ranks candidate policies as the target environment would. Table 1 summarises the input–output contract of each stage; Appendix Algorithm 1 states the simulation loop in full.

3.2 Patients: vignettes, personas, and synthetic populations

A patient enters the simulator as a structured vignette, consisting of demographic and behavioural fields read from the study records, paired with a persona summarising how they tend to respond. Structured-vignette conditioning follows the patient-vignette and simulated-patient literature [26,3,31]; the persona adds a compact description of response tendencies. At each decision point the model is given the vignette, the current context (for HeartSteps: time of day, location, weather, and the action taken) and the trajectory simulated so far, and predicts the next outcome; an example vignette and the prompt assembled from it, with the participant’s fields substituted into the template slots, are shown in Appendix Figure A1 and Figure 2, respectively.

Prompt — HeartSteps step response

heartsteps-context-faithful

System You are a calibration-focused behavioural forecaster predicting one step of a physical-activity JITAI. Estimate what an external monitor would record for this decision point, given the participant's profile, prior history, current context and current intervention. The outcome is execution behaviour, not intent: concrete contextual friction (poor location, weather, transit, fatigue) outweighs positive traits or a supportive prompt, and a single intervention rarely flips a low-outcome step into a high one. Preserve intermittency; the current outcome is hidden, so infer it and never fabricate it.

Profile female, 48, some college, program coordinator; exercise self-efficacy 19 (high), conscientiousness 21 (high), walking at intake 4.

Base-rate priors [Nahum-Shani et al. 2018 and behavioural-science priors — single-prompt effects are small and context-dominated; elided.]

Context study day 4, morning (slot 1); location home; weather Clear, 16°C; steps in prior 30min 0; hours since phone use 4.7.

Warm-start prefix the participant's first $K = 14$ steps with their real, measured outcomes — anchors the model in how this person actually behaves

step 1 [day 1, afternoon]: sedentary_break_suggestion → 0.632

step 2 [day 1, evening]: active_suggestion → 0.000

: real outcomes through step 14

Recent simulated history up to the most recent seven post-anchor predictions are fed back; the first post-anchor prompt has none

recent step: active_suggestion → 0.150 (predicted)

: rolling history window, most recent last

Current action (observed trial action) one of no_suggestion, active_suggestion, sedentary_break_suggestion.

Return strict JSON {"adherence": 0-1, "adherence_certainty": 0-1, "reason_code": ..., "state_update": {...}}

Fig. 2. Schematic condensation of the $K=14$ HeartSteps context-faithful prompt (shown for participant 1 and an observed trial action; coloured values are filled in at run time, while detailed priors and guidance are elided). The field names and seven-step history window match the experiment configuration. The model returns strict JSON; the emission layer then turns the predicted adherence value into the realised outcome (Figure 1).

Benchmarking policies requires a larger population than the few dozen real patients, so the framework samples structured synthetic profiles from predefined ranges of age, baseline activity, prompt reactivity, and habituation. The language model then generates each profile's response trajectory one day at a time.

3.3 The LLM behavioural model

At the core of the simulator a language model predicts a latent expected tendency $\ell_t \in [0, 1]$ from the vignette, persona, context and simulated history, entirely zero-shot (prompted, never fine-tuned), so the only component ever trained on data is the RL policy fitted later inside the finished simulator. The model is framed as a calibration-focused, *sceptical forecaster* rather than an enthusiastic coach and asked to accompany each prediction with a short reason code; both choices are deliberate, because a naive first-person framing collapses the differences between patients (Section 4.1). Two optional prompt mechanisms target

the model’s tendency to overstate how strongly interventions help. The first is a *stateful* variant in which the model maintains a compact internal state z_t , a small set of named latent factors such as stress, fatigue, or motivation momentum, returned updated alongside each prediction and carried across steps:

$$z_t = \text{LLM}_{\text{state}}(z_{t-1}, c_t, a_t, y_{t-1}), \quad \ell_t = \text{LLM}(c_t, a_t, P, A_{1:K}, H_{K+1:t-1}, z_t), \quad (3)$$

with z_0 initialised to a neutral value. Equation (3) conditions ℓ_t on z_t , letting the simulator express the slow within-patient dynamics associated with the habituation and disengagement (“cognitive drop-off”) by which repeated prompting loses effect over time, which the memoryless map of Equation (1) cannot. The second is an *action-noise hint* instructing the model that intervention effects are small and noisy and must not be applied as a uniform positive shift. Both are evaluated rather than assumed beneficial (Section 4.1).

3.4 Warm-start anchoring

In the real-data validation experiments, each simulated participant is grounded in the first K decision points of their real trajectory, used only as conditioning and to set a patient-level baseline; later outcomes are never read, so the simulator is always scored on a period it has not seen. The synthetic RL populations have no real prefix and instead self-anchor on their generated outcomes. On HeartSteps (up to five decision points per day) the default $K=35$ is roughly one week of observation. The anchor length $K \in \{0, 14, 35, 70\}$ is treated as an experimental variable; shrinking K emulates a new or barely observed user.

3.5 The emission layer

A language model used on its own produces trajectories far too smooth to resemble real behaviour. Their within-patient variability is only about a quarter of the real value (Section 4.1), so rather than emit the outcome directly we pass the latent tendency through a statistical emission layer that depends only on information available up to the current step (so it remains causal) and carries no dataset-specific parameters. Behavioural-intervention outcomes often combine zero observations with a continuous range of positive values. The layer therefore uses a Cragg-style two-part, or hurdle, model [5]: one part models whether a response occurs and the other its magnitude, while the language model supplies the expected response. Writing $\sigma(\cdot)$ for the logistic (sigmoid) function and $\text{logit}(\cdot) = \sigma^{-1}(\cdot)$ for its inverse, the layer first decides whether the response is positive and, if so, draws its magnitude from a log-normal distribution:

$$p_t = \sigma((1-w)\text{logit}(\ell_t) + w\text{logit}(r_b)), \quad (4)$$

$$m_t = \text{clip}_{[0.005, 1]} \left(\frac{(1-w)\ell_t + w\bar{b}}{p_t} \right), \quad (5)$$

$$y_t = \begin{cases} 0 & \text{with probability } 1 - p_t, \\ \text{clip}(\text{LogNormal}(\log m_t - \frac{1}{2}\nu^2, \nu)) & \text{with probability } p_t, \end{cases} \quad (6)$$



Fig. 3. The two-part (hurdle) emission, Equations (4)–(6). The clipped LLM tendency is blended with the patient’s anchor positive-response rate to set the Bernoulli gate p_t and with the anchor mean to set the target mean; dividing by p_t gives the open-gate conditional mean m_t . A closed gate gives $y_t=0$; an open gate draws a clipped log-normal magnitude with spread ν . The aligned panels show the first fourteen simulated days of a schematic example.

Here, $r_b, \bar{b} \in [0, 1]$ are the positive-response rate and mean of the warm-start anchor outcomes $A_{1:K}$, $w \in [0, 1]$ weights the patient baseline against the model prediction, p_t is the hurdle (probability of a positive outcome), m_t the conditional mean of the positive part, $\text{clip}(\cdot)$ restricts to its displayed interval, and ν is the log-normal dispersion; the offset $-\frac{1}{2}\nu^2$ centres the log-normal so that its expectation equals m_t . After the first simulated step a mild persistence adjustment nudges the hurdle toward the previous simulated outcome ($p_t \leftarrow 0.9 p_t$ after a zero; $p_t \leftarrow 1 - 0.9(1 - p_t)$ after a clearly active step, $y_{t-1} \geq 0.25$; then clipped to $[0.01, 0.99]$); these two fixed constants inject a small lag-one dependence, which is partly responsible for the temporal-persistence match reported later, and are held fixed alongside the two free parameters. We do not tune this layer to the data: ν and w are fixed at 0.9 and 0.5 throughout, and their influence is reported separately as a sensitivity check rather than optimised. Where positive responses are common the hurdle stays open ($p_t \rightarrow 1$) and the emission behaves as a continuous distribution. Figure 3 illustrates the mechanism and the outcome distribution it produces.

3.6 Evaluation methodology

The choice of evaluation is a central commitment: a simulator intended for policy training must satisfy three properties, each measured by an explicit metric motivated by prior work and stated as a comparison of a simulated quantity against its real counterpart (Table 2).

Table 2. The evaluation measures: what each establishes, what it does not, and the prior work motivating it (Section 3.6).

Measure	Establishes	Does not establish	Basis
Distinguishability (C2ST)	selected-feature similarity	raw level, action effect, decisions	[19]
Patient ranking (Spearman)	between-patient ordering	within-patient dynamics	operationalises [2]
Heterogeneity ratio	between-patient spread	within-patient variance	operationalises [2]
Temporal persistence (autocorr.)	lag-one dynamics	dispersion, action effect	this work
Marginal action effect	proximal action-response fidelity	marginal realism	[14]
Policy transfer and ranking	downstream policy value	distributional realism	[8,21]

1. **Distributional realism.** Simulated trajectories should be shaped like real behaviour rather than smooth or degenerate, measured with the classifier two-sample test (C2ST), the standard two-sample diagnostic for generative models [19]: a classifier is trained to separate real from simulated trajectory features and its cross-validated accuracy reported, complemented by lag-one autocorrelation as a direct temporal-persistence diagnostic.
2. **Between-patient structure.** Motivated by Argyle et al.’s broader “algorithmic fidelity” criterion [2], we operationalise population structure as the Spearman rank correlation between simulated and real patient means and a heterogeneity ratio of their spreads.
3. **Proximal action response.** What a policy learns from is how the outcome changes when an intervention is delivered. For each suggestion type a , we estimate the marginal contrast $\Delta_a = \mathbb{E}[y \mid a] - \mathbb{E}[y \mid \text{no_suggestion}]$ in both the real and simulated trajectories, with 95% intervals obtained by bootstrapping patients, and compare the simulator’s action response against the micro-randomised trial [14]. The downstream measures are policy transfer, following the world-model principle of Ha and Schmidhuber [8], and policy-ranking fidelity, an extension introduced here; both align with the decision-focused analysis of Martinson et al. [21].

In this implementation, distributional realism is a summary-feature diagnostic: the C2ST measures distinguishability with respect to the selected features and classifier, not whether decisions taken on the trajectories are sound. Policy transfer and ranking fidelity therefore provide the downstream test.

We deliberately do not judge the simulator by how accurately it predicts any single day, which is dominated by unobserved events and noise. The C2ST uses transferable summary features of each trajectory (variability, lag-one autocorrelation, volatility, rate of large day-to-day changes, average response under each action), five-fold cross-validation split by patient, and mean-shifted outcomes, so it scores trajectory shape rather than level or memorised individuals; accuracy runs from 0.5 (indistinguishable, the ideal) to 1.0, with resolution about 1/37 on the thirty-seven-patient sample, so small differences are treated as ties.

Statistical reporting. Principal stochastic results are reported with uncertainty, and significance is tested where a defined reference exists; Appendix A.5 gives the complete reporting protocol.

3.7 Datasets

The empirical evaluation uses three datasets. The primary testbed is HeartSteps, a public micro-randomised trial in which activity suggestions were delivered at repeated phone decision points and the proximal thirty-minute step count was recorded [15]. We normalise this outcome to the unit interval ourselves (dividing by fifteen hundred and clipping) and use the thirty-seven participants with full trajectories averaging about 165 decision points; after the $K=35$ warm-start prefix, the scored windows average about 130. The REINFORCE trial’s day-level records, the project’s original target [16], were predominantly zero even where aggregate adherence was high, making them unsuitable as a reference distribution. The published HeartSteps analysis found a modest effect that attenuated over the study [15]; under the marginal estimator and preprocessing used here, the observed contrast is effectively zero (Section 4.2). This provides the weak-effect test, while StepCountJITAI covers a strong known effect. To test whether the framework is specific to HeartSteps we also use HPTN 067 [7], a pre-exposure-prophylaxis adherence trial whose weekly coverage outcome is high-valued and not zero-inflated in the data used here. Finally, because neither real dataset has a known intervention effect against which transfer can be measured, we use StepCountJITAI, a synthetic activity-intervention environment [11]. Its latent state $[c_t, h_t, d_t]$ comprises context, habituation and disengagement; four actions cover no message, a generic message and two context-specific messages, with higher activity for a correctly matched message and penalties for over-messaging. The parameter σ controls error in the inferred context and σ_s controls activity noise. Learned policies use the inferred context and a capped consecutive-message count because a deployed mobile intervention can infer context and track its consecutive nonzero messages. True context, habituation and disengagement are simulator-internal states that a mobile app does not directly record; exposing them would give a learned policy information unavailable at deployment, so they are reserved for the full-state oracle.

Unless stated otherwise the language model is Qwen3.5-122B (A10B, NVFP4), served on the Nebula platform of the Network Institute at Vrije Universiteit Amsterdam, queried at temperature 0.6 with nucleus sampling and a fixed seed; the model-capability experiment also uses two commercial models.

3.8 Experimental design

The experiments follow the logic of the evaluation methodology, moving from realism to action response to downstream decisions; Appendix Table A1 states each experiment as a question, the manipulation that answers it, the measures read off, and the research question it serves.

The reinforcement-learning testbed is an autoregressive day-by-day LLM environment. At each decision point, one model call receives the patient profile, realised history, current context and selected action, and returns the activity tendency, a disengagement flag and the next context; the emission layer is *self-anchored* on the patient’s generated outcomes. The learner is tabular Q -learning

(state = inferred context $\times$ capped consecutive-message count, four actions). Offline learning fits the policy to three fixed corpora of simulated trajectories, whereas online learning grows its training data through actions selected by the current policy. After training, each policy is frozen and evaluated in StepCountJITAI. We compare it with the same Q -learning class trained directly from the observable state (the *direct-training reference*) and with a privileged $[c, h, d]$ state (the *full-state oracle*, which also cross-checks the reward scale against the environment’s authors).

Reproducibility. Code, prompts, prepared data, saved results, seeds, model identifiers and decoding settings are available at <https://github.com/Nolan064/llm-sim-rl-health>. The quantitative analyses can be rerun from the saved artifacts on CPU.

4 Results

The results follow the validation ladder of Section 3.6: distributional realism first (Section 4.1), then the proximal action response (Section 4.2), generalisation and necessity checks (Section 4.3), and finally the policy-transfer and benchmark-ranking tests (Section 4.4).

4.1 The realism gap and its causes (RQ1)

Conditioning the model on a patient’s own warm-start history is necessary for realism. Without any anchor the simulator cannot rank patients (Spearman 0.04) and inflates the average response by roughly a factor of two (Appendix Table A2, first row). A short warm-start ($K=14$ decision points, about three days) is sufficient to recover patient ranking, mean calibration and temporal persistence. Longer histories add little: classifier accuracy remains near 0.97 at every anchor length (Figure 4, top rows), showing that the remaining realism gap arises elsewhere. The feature-group analysis points to two sources. Simulated trajectories are too smooth, with within-patient variability about one quarter of the real value. They also show a stronger and more consistent action response than the real data: removing action-related features produces the largest reduction in classifier accuracy.

Prompt, model, and target ablations. The ablations examine three possible explanations for the residual gap: model capability, prompt content and output format. Figure 4 summarises the ablations; the full tables are in the Appendix (Tables A3–A5). Model capability matters: on the matched twelve-patient subset, GPT-5.4 is less separable than Qwen3.5-122B (0.875 versus 1.000) and GPT-5.4-mini (0.958). The full-sample Qwen3.5-122B, DeepSeek-V4-pro, gpt-oss-120b and Gemma3-12B runs remain highly distinguishable (0.959–0.986), so the stronger model narrows but does not close the realism gap.

Removing the domain description, persona, weather and location, intervention text, or all prose at once leaves C2ST accuracy at 0.973 (Appendix Table A4). A naive first-person framing is the exception: realism is unchanged, but patient ranking collapses (Spearman $0.58 \rightarrow 0.04$) and heterogeneity falls to near zero. A history-only prompt retains ranking, so numeric warm-start history carries most measured patient-specific signal for anchored HeartSteps participants. Profiles serve the cold-start RL population: before a synthetic rollout has any history, sampled baseline activity, prompt reactivity and habituation are its only patient-specific inputs. Changing the output format does not help: discrete behaviour-class targets give at best a negligible improvement while predicting the intended class near chance. The largest improvement comes from the emission layer.

The emission layer. Adding the online two-part emission layer lowers the classifier accuracy from about 0.97 to about 0.74 (recall that 0.5 is indistinguishable) and moves within-patient variability toward the real-data value (Figure 4, blue rows; per-configuration values in Appendix Table A6). The exaggerated action response is more difficult to remove. We evaluated the stateful variant and the action-noise hint (the two prompt mechanisms of Section 3.3) as a 2×2 factorial on top of the two-part layer. Across the four combinations, C2ST accuracy ranges from 0.716 to approximately 0.76. Because these estimates differ by less than one cross-fold standard deviation, we treat the prompt variants as tied and attribute their common improvement to the two-part layer. Even the best configuration leaves the classifier separating the two sets most of the time, because the heavily zero-inflated tail is not reproduced cell by cell: the simulator is a measurable improvement on a prompt-only model, but remains distinguishable from real data.

4.2 The simulator overstates the intervention effect (RQ2)

The third evaluation criterion of Section 3.6 is the proximal action response. Using the marginal contrast defined above, the observed HeartSteps effect of active suggestions is effectively zero (-0.0004 ; 95% bootstrap CI $[-0.014, +0.013]$). The language model nonetheless produces a positive effect of about $+0.036$ for active suggestions (95% bootstrap CI $[+0.021, +0.052]$) and $+0.014$ for sedentary suggestions, substantially larger than the effects observed in the data (Figure 5).

The stateful variant and action-noise hint of Section 3.3 reduce the simulated contrast from $+0.036$ to $+0.017$, although it remains above the observed contrast.

4.3 Generalisation and necessity checks

We probe the construction of the framework by examining what the language model adds beyond the emission layer, whether the pipeline is specific to HeartSteps, and whether its fixed parameters were tuned to the data. These checks continue to use trajectory-feature C2ST accuracy (0.5 indistinguishable; the headline HeartSteps configuration scores approximately 0.72).

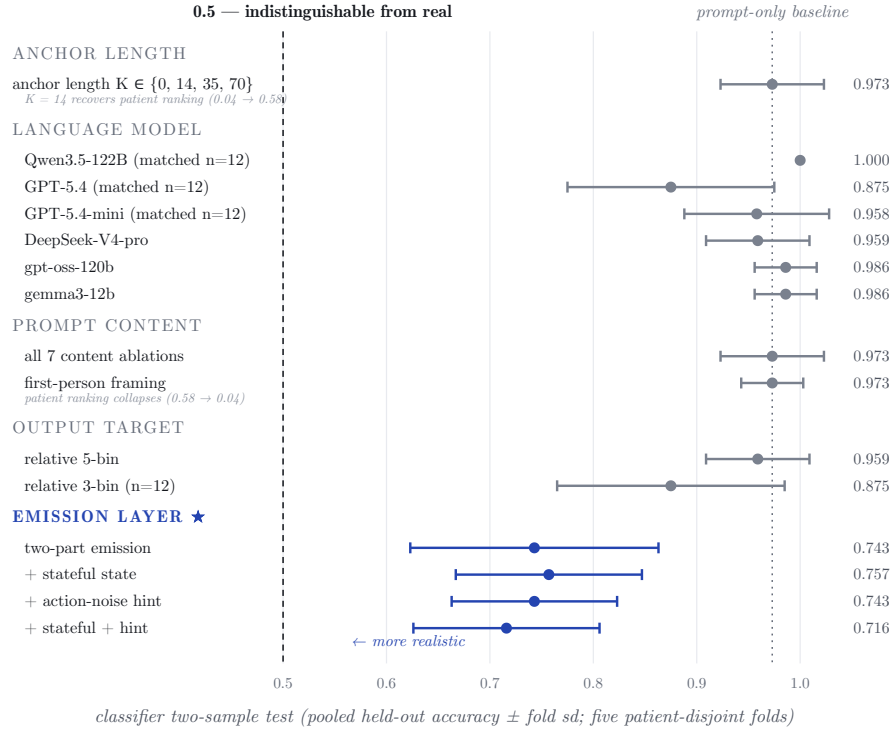


Fig. 4. Only the emission layer materially improves realism. One-figure summary of the ablations (RQ1; full tables in Appendix Tables A2–A6): dashed line 0.5 = indistinguishable, dotted line = prompt-only baseline. Labels marked “matched $n=12$ ” use the same twelve patients for the commercial-model comparison. No anchor length, model substitution, prompt content or output format (grey) approaches chance. The emission-layer variants (blue) are the only substantial movement (0.973 $\rightarrow$ 0.72–0.76). The one consequential prompt change is the first-person framing: realism is unchanged, but patient ranking collapses (Spearman 0.58 $\rightarrow$ 0.04) and heterogeneity falls to near zero.

Contribution of the LLM beyond the emission layer. This is a distributional component ablation only. Replacing the LLM with each patient’s fixed anchor mean removes any action response, so the control cannot serve as an RL training environment; its purpose is to separate the marginal realism supplied by anchoring and emission from the dynamics supplied by the language model (Appendix Table A7). At the full anchor ($K=35$), the control matches the LLM’s patient ranking (Spearman 0.648 vs. 0.647) and is less distinguishable (C2ST 0.541 vs. 0.743). With little history, the LLM contributes more: at $K=2$ it preserves ranking better (0.35 vs. 0.26) and keeps the mean closer to the observed 0.107 (0.106 vs. 0.130).

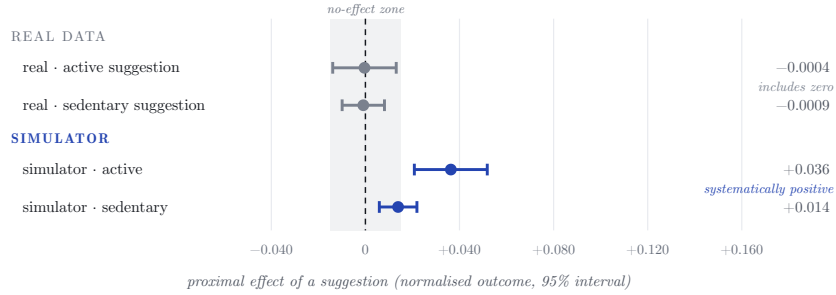


Fig. 5. Observed and simulated intervention effects on HeartSteps (dashed line marks no effect). The real proximal effects (grey) straddle zero, whereas the simulated action effects (blue) lie entirely above it.

A second dataset. We apply the identical emission layer, unchanged and un-tuned, to HPTN067, a PrEP adherence trial whose weekly coverage outcome is high-valued and not zero-inflated and therefore differs structurally from HeartSteps. With an eight-observation warm-start, 87 of 96 patients provide post-anchor outcomes; 77 have the minimum five observations needed for trajectory-feature C2ST. This is a distributional test rather than an RL environment because HPTN assigns a regimen once instead of choosing interventions repeatedly. The LLM-driven layer reaches C2ST accuracy 0.753, compared with 0.695 for the anchor-only control, while preserving patient ordering (Spearman 0.55) and much of the between-patient spread (heterogeneity 0.77). A uniformly positive LLM bias might be expected, but HPTN goes in the opposite direction: its high observed mean is pulled down from 0.68 to 0.39. Taken with the upward error on HeartSteps, the two directions are consistent with a central-tendency (moderation) bias, in which extreme values are pulled toward the middle, rather than a uniformly positive bias, although two datasets are not enough to establish this as a general rule.

A tuning-sensitivity check. Because the emission parameters (ν, w) are fixed a priori rather than tuned, we vary both around their defaults (Appendix Table A8). Changing the anchor weight w has the larger effect: giving the anchor more weight improves distributional similarity (C2ST accuracy $0.93 \rightarrow 0.62$) while slightly reducing patient-ranking correlation ($0.66 \rightarrow 0.56$) and increasing between-patient spread toward and then beyond the real value (heterogeneity $0.77 \rightarrow 1.11$). Changing the dispersion parameter ν changes accuracy by less than one across-fold standard deviation at either anchor weight.

4.4 Benchmarking RL policies: train on synthetic, test on the reference environment (RQ3)

The final experiments test two uses of the simulator: transfer, whether a policy trained inside it performs well on StepCountJITAI, and benchmark validity, whether simulator scores reproduce StepCountJITAI’s ranking of candidate policies. Throughout this subsection, J denotes return measured in StepCountJITAI. The learner is tabular Q -learning [29], kept deliberately simple since the object under test is the simulator, not the RL algorithm; every policy observes only what a deployable controller could see: the inferred context c_t^{inf} and its capped consecutive-message count, never the latent habituation h_t or disengagement d_t . Results are reported on StepCountJITAI’s paper-parameter-aligned configuration ($\sigma=2$, $\sigma_s=20$), corresponding to approximately 40% context-inference error; sensitivity down to a clean $\sigma=0.4$ (approximately 10% error) is reported alongside (Figure 7) [11].

A policy trained in the simulator transfers to the reference target. A policy trained entirely on the synthetic population transfers to the reference environment, whether trained offline on fixed corpora of 400 sampled profiles with LLM-generated trajectories or online with the policy choosing each day’s action and the model simulating the response (Table 3). The online policy reaches 1013 ± 70 , well above the random floor (575) and the no-intervention floor (22), and recovers 94% of the direct-training reference (1080) — the return of the same Q -learning policy class trained directly on the reference environment with abundant interaction. This is achieved despite training inside the LLM with roughly an order of magnitude less data and without contact with that environment. The offline variant is markedly weaker and more variable (667 ± 273 , median 776): batch Q -learning from a fixed corpus occasionally collapses to the no-action policy; one of nine fits selected no action throughout, illustrating sensitivity to a fixed batch [17]. The online policy also exceeds the always-context-matched heuristic by 232 return (1013 vs. 781), showing that direct context matching alone does not account for its transfer performance.

Validation against the reference environment, and the cost of partial observability. The full-state oracle reaches 3235, close to the approximately 3000 reported for DQN, PPO and REINFORCE with privileged $[c, h, d]$ observations, confirming the implementation and reward scale [11]. The large difference between the deployable direct-training reference (1080) and the oracle quantifies the cost of acting without the latent state. The isolated context-inference sweep below then tests how one component of that partial observability affects policy ranking.

Benchmark validity. We next test whether the simulator’s verdict on a policy agrees with the reference environment by comparing the rankings induced by J and true return. We evaluate 49 deployable policies spanning the full quality range, including fixed and probabilistic message rules, a heuristic-intensity sweep, and the offline and online learned policies and their blends, under one

Table 3. Train-on-synthetic, test-on-reference transfer on the paper-parameter-aligned StepCountJITAI ($\sigma=2$). Deployment policies observe inferred context and a capped consecutive-message count; the full-state oracle is a privileged reference. $\dagger$ reference lines: the direct-training reference is the same Q -learning class trained on the reference environment itself; the full-state oracle additionally reads h, d .

Method	Trains on	Test return
no intervention (floor)	—	22
random	—	575
heuristic (always context-matched)	— (hand-coded)	781
offline Q -learning	synthetic LLM population	667 ± 273
online Q -learning	synthetic LLM population	1013 ± 70
direct-training reference †	reference environment	1080
full-state oracle † (reads h, d)	reference environment	3235

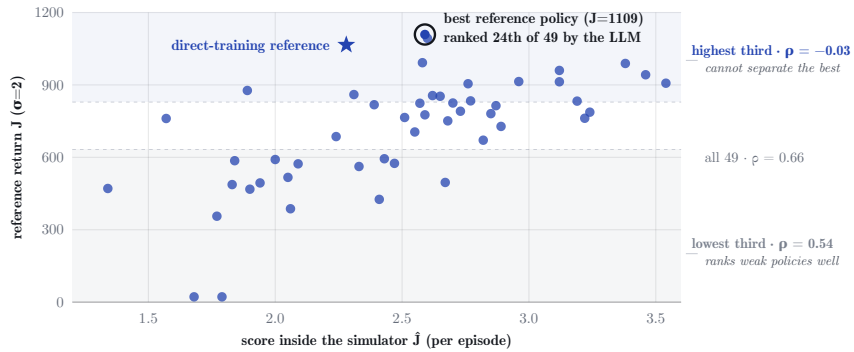


Fig. 6. Benchmark validity: LLM-world score $\hat{J}$ against the reference environment's true return J ($\sigma=2$) for 49 deployable policies. The rank agreement ($\rho=0.66$) is concentrated in the lowest third by reference return; in the highest third the LLM score is essentially flat; the true best policy ($J=1109$) is mid-range in $\hat{J}$. The direct-training reference (star) sits mid-pack.

prompt on the same simulated patients, each policy scored over three independent patient batches (Figure 6; per-family detail in Appendix Table A9). The overall agreement is **moderate and significant**: Spearman $\rho=0.66$ (95% bootstrap CI [0.45, 0.81]; per-batch $\rho=0.59 \pm 0.03$).

The aggregate correlation varies substantially across the quality scale. Splitting the policies into thirds by true return, rank agreement is good in the lowest third ($\rho=0.54$) but essentially zero in the highest ($\rho=-0.03$): the LLM's top five and the true top five share only one policy, and the single best policy on the reference target (the online-trained Q -learner, $J=1109$) is ranked 24th of 49. Choosing the simulator's top-ranked policy gives $J=907$, which is 18% below 1109. Even allowing a choice from its top-five shortlist, the best policy in that shortlist reaches 989, an 11% loss. The top-ranked choice nevertheless clears the

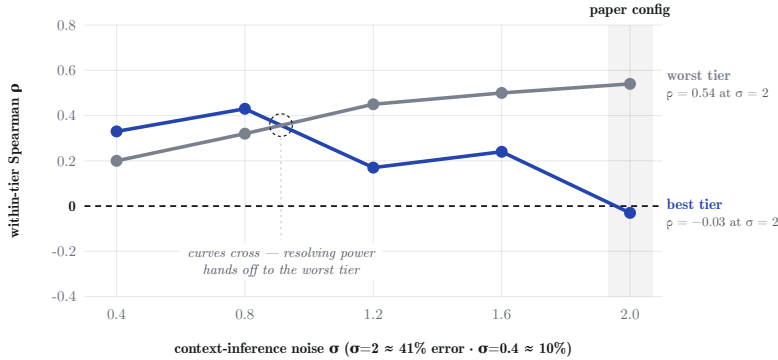


Fig. 7. Benchmark resolution vs. context-inference noise σ (all else fixed; paper config $\sigma=2$). With cleaner inference (left) the simulator resolves the best policies, placing the true best 3rd–8th of 49 against 24th at $\sigma=2$; with noisier inference (right) resolution shifts toward the worst policies, consistent with context-observation mismatch contributing to the ranking gap.

median policy ($J=765$) and the hand-coded heuristic (781) comfortably. The simulator therefore provides useful screening: the moderate ρ reflects its separation of obviously poor policies from competent ones, while its scores do not resolve the best policies.

The errors are systematic: the simulator over-rates idle and over-messaging policies (a do-nothing policy, worth $J=22$, receives a respectable LLM score from baseline activity; the most aggressive heuristic mixtures reach $\hat{J}$ -rank 4–5 yet sit mid-pack) and under-rates the learned policies that are in fact best.

Sensitivity to inference noise. Re-scoring the same policies across context-inference settings changes where the simulator has ranking resolution (Figure 7). With lower inference noise, agreement among the best policies improves and the true best policy rises to third–eighth place; at the paper-aligned $\sigma=2$, it ranks 24th. This improvement is consistent with context-observation mismatch contributing to the ranking gap, but the sweep does not isolate it from other action-response errors.

5 Discussion

Simulation fidelity (RQ1–2). Anchoring and the two-part emission layer bring the classifier accuracy from 0.97 down to 0.72. This is a measurable improvement in trajectory shape, although not indistinguishability, and that level of fidelity is already enough to train a policy that recovers 94% of the direct-training reference: the simulator becomes useful for training before it becomes realistic. A no-LLM control locates where the model earns its place: with a full anchor the anchors and the emission layer carry marginal realism on their own, while

the LLM’s remaining contribution is the action response and a modest, monotone advantage on under-observed patients (Section 4.3). The decision-relevant structure is captured only in part: between-patient ordering is preserved once anchored (Spearman ≈ 0.6), but the model produces a $+0.036$ intervention contrast where the observed contrast is effectively zero. The prompt mechanisms reduce it to $+0.017$ without removing it, a limit that matters where proximal effects are typically near zero.

Policy transfer and benchmark validity (RQ3). Transfer and policy selection fare differently. A policy trained entirely inside the LLM world recovers 94% of the direct-training reference with an order of magnitude less data and no contact with the target; but the simulator’s ranking of 49 policies is only moderately correlated with the reference environment ($\rho=0.66$), and that correlation reflects only coarse resolution: agreement is good in the lowest third ($\rho=0.54$) but absent in the highest ($\rho=-0.03$), leaving the true best policy unidentified. The errors are systematic: the LLM world over-rates idle and over-messaging policies and under-rates the best learned policies. Partial observability and the context-observation mismatch contribute to both deployable transfer and the benchmark’s ability to select. The simulator still preserves the coarse advantage of context-matched messaging, which is sufficient for the learned policy to transfer. These results show that training and ranking are separate abilities: a simulator can produce one transferable policy without reliably selecting the best among many. One scope note: all RQ3 evidence comes from StepCountJITAI; the real-data evidence of HeartSteps and HPTN067 enters through RQ1–2, and the two strands should not be read interchangeably.

Design implications. The small discrete state–action space makes tabular Q -learning [29] a transparent fit and keeps learner variance from obscuring simulator error. The testbeds use two forms of history. Realism experiments condition on each participant’s first K observations. In RL rollouts no real prefix is used; after the first step, each patient’s earlier simulated outcomes form the emission layer’s evolving baseline. The behavioural model is used zero-shot, so new profiles and intervention semantics are introduced without task-specific fine-tuning; carried state and reason codes also make individual steps inspectable (Section 3.3).

Limitations. The full itemised list is in the Appendix. Transfer and ranking use one synthetic reference target, and the HPTN067 evidence is distributional only. The HeartSteps effect is a marginal, patient-bootstrapped action contrast. StepCountJITAI is structurally simple, with two contexts, four actions, and hand-specified dynamics, so transfer in this controlled testbed may overstate performance in richer clinical environments. Downstream results use one model and three online training seeds, and the reference values are finite-run estimates. The best simulator remains distinguishable and moderately biased; on the hard target it screens poor policies but lacks top-tier resolution, with noisy per-policy scores. These boundaries mark where the numerical estimates are indicative.

6 Conclusion

This thesis asked whether an LLM-based synthetic-patient simulator can provide a reliable benchmarking environment for RL policies targeting personalised adaptive health interventions. **RQ1**: with warm-start anchoring and the emission layer the simulator approaches, but never reaches, distributional realism (C2ST $0.97 \rightarrow 0.72$); neither model substitution nor prompt ablation eliminates the residual gap. **RQ2**: between-patient ordering is preserved, but the simulator produces a positive intervention contrast absent from the data; prompt-level mitigations reduce it without removing it. **RQ3**: a policy trained entirely inside the simulator transfers, recovering 94% of the return of the same learner trained on the known-effect reference environment, while the simulator ranks 49 policies moderately ($\rho=0.66$) but coarsely. It reliably rejects poor policies yet offers little signal among the best; cleaner context inference improves, but does not eliminate, that gap. The overall answer is a qualified yes: an LLM simulator can train a deployable policy and coarsely screen candidates even when its trajectories remain distinguishable from real data. These findings support validation through action-response fidelity, policy transfer, and ranking fidelity, because marginal realism alone can miss the action-response and ranking failures documented here. Residual distinguishability and over-stated near-zero effects should therefore be treated as explicit limits on use.

Future work. The present results motivate work on validation targets, model calibration, and clinical evidence.

Validation targets. A semi-synthetic HeartSteps environment with prespecified proximal effects would retain observed patient heterogeneity and outcome noise while providing known ground truth. Repeating the transfer and ranking experiments across effect sizes, policy families and environments with richer context and habituation would show how broadly the findings generalise. Policy selection could also be evaluated through the true return of simulator-chosen policies and action-specific contrasts, alongside rank correlation.

Model and calibration. Repeating the benchmarks with stronger LLMs would show how much fidelity depends on model capability. A direct-LLM policy using the same deployable observations would provide a useful comparison with simulator-trained Q -learning [13]. Calibration work could explore shrinkage of simulated action contrasts and aggregate fine-tuning across health-behaviour datasets, while tracking realism, action-response fidelity, policy transfer and inference cost.

Deployment and interpretation. A deployment-oriented simulator should infer context, habituation and disengagement from app-observable histories and be tested under noisy or missing state estimates. Following retrospective validation, the system could progress to silent prospective evaluation and then a small real-patient study that also records burden, engagement and message acceptance. The model’s carried state and reason codes could be examined across prompts and models to understand simulation errors and guide further refinement [12].

Declaration of AI assistance technologies during manuscript preparation. During preparation of this manuscript, the author used generative AI tools for language and grammar correction, LaTeX formatting, assistance with developing and debugging parts of the experiment and analysis code, and generating and formatting figures and tables from verified results. The author reviewed and edited all outputs, verified the reported results against the experiment logs, and takes full responsibility for the manuscript. Large language models are also an object of study in this work; that experimental use is described separately in the methodology.

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Appendix

This appendix contains the simulator pseudocode (Algorithm 1); a worked vignette example (Figure A1); the experimental programme (Table A1); and the supporting ablation, sensitivity and benchmark tables (Tables A2–A9). It also gives the complete statistical-reporting protocol and the full itemised list of limitations.

A.1 The simulation loop, and worked examples of the simulator’s inputs

Algorithm 1 The simulator loop: one patient trajectory under policy π (Figure 1). The simulator score $\hat{J}(\pi)$ averages the returned totals over simulated patients.

Require: profile $P = (V, \rho)$; warm-start anchor $A_{1:K}$ with statistics $(r_b, \bar{b})$; policy π ; horizon T ; emission parameters (w, ν)

- 1: $H \leftarrow A_{1:K}$ $\triangleright$ history starts as the K real anchor outcomes
- 2: **for** $t \leftarrow K+1$ **to** T **do**
- 3: observe context c_t ; assemble the deployable state s_t from c_t and the recent history H
- 4: $a_t \leftarrow \pi(s_t)$ $\triangleright$ the policy under training or evaluation
- 5: $\ell_t \leftarrow \text{LLM}(c_t, a_t, P, A_{1:K}, H_{K+1:t-1})$ $\triangleright$ zero-shot latent tendency, Eq. (1)
- 6: $y_t \sim \mathcal{E}(\ell_t; r_b, \bar{b})$ $\triangleright$ two-part emission, Eqs. (4)–(6)
- 7: append y_t to H $\triangleright$ autoregressive: simulated outcomes condition later steps
- 8: **end for**
- 9: **return** trajectory $(y_{K+1}, \dots, y_T)$ and its return $\sum_{t=K+1}^T y_t$

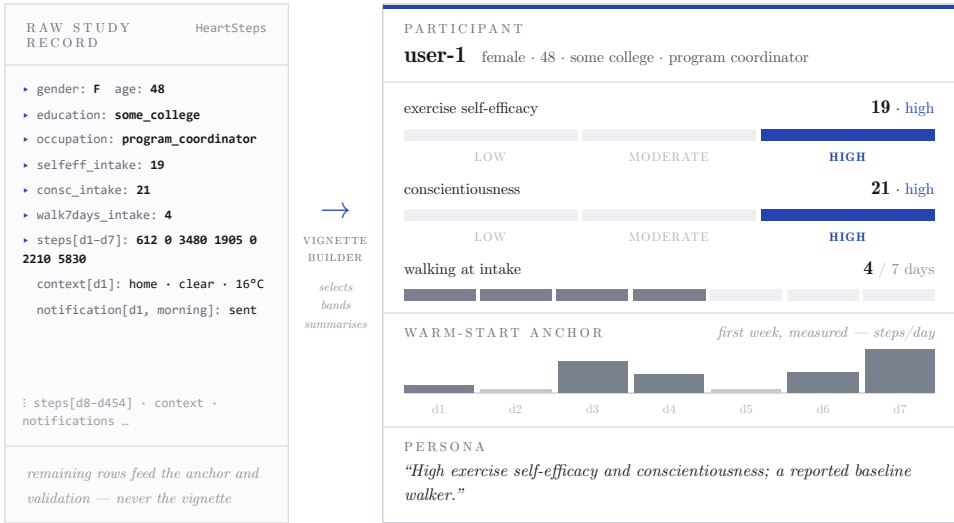


Fig. A1. An example HeartSteps participant vignette (participant 1): the raw study record (left) is rendered into the patient profile the simulator conditions on (right) — demographic header, behavioural trait scores, the measured first week that seeds the warm-start anchor, and the free-text persona (italics). The fields are read from the study records (Section 3.7); for the synthetic populations the profile fields are sampled from predefined trait ranges and assembled into structured persona strings.

A.2 The experimental programme

Table A1. The experimental programme: question, manipulation, measures, and the research question served; results follow the same order in Section 4.

Question	Manipulation / setup	RQ	Results
Are the trajectories realistic, and what governs realism?	anchor length $K \in \{0, 14, 35, 70\}$; six models under a fixed prompt; prompt-content ablations; discrete-target reformulation; emission layer with a 2×2 prompt factorial	RQ1	§4.1
Does the simulator reproduce the proximal action response?	marginal action contrast in real and simulated trajectories with bootstrap confidence intervals	RQ2	§4.2
Does the framework generalise, and what does the LLM add?	LLM replaced by the patient's anchor mean (no-LLM baseline) at each anchor length; identical layer, untuned, on HPTN 067; (ν, w) sensitivity sweep	RQ1–2	§4.3
Does a trained policy work on the reference target, and does the simulator rank policies as it does?	train inside the LLM world, freeze, test on StepCountJITAI (never trained on); 49-policy benchmark over three patient batches; direct-training and oracle references; five context-noise settings ($\sigma \in \{0.4, 0.8, 1.2, 1.6, 2.0\}$)	RQ3	§4.4

A.3 Full ablation tables

Table A2. Prompt-only HeartSteps baselines as a function of anchor length K (full $n=37$). Anchoring restores ranking, calibration and persistence, but not distinguishability.

K	C2ST acc.	Spearman	autocorr. err.	mean sim/real
0	0.973 ± 0.05	0.039 ± 0.17	0.379 ± 0.04	0.200/0.107
14	0.973 ± 0.05	0.560 ± 0.13	0.135 ± 0.01	0.093/0.106
35	0.973 ± 0.05	0.579 ± 0.12	0.119 ± 0.01	0.098/0.104
70	0.986 ± 0.03	0.573 ± 0.15	0.145 ± 0.02	0.092/0.104

Table A3. Prompt-only comparison across models ($K=35$). The three twelve-patient rows use the same patient subset, so the GPT-5.4-versus-Qwen comparison is matched; capability helps but does not by itself solve the realism gap.

Model	n	C2ST acc.	mean sim/real
Qwen3.5-122B	37	0.973 ± 0.05	0.098/0.104
Qwen3.5-122B (same 12)	12	1.000 ± 0.00	0.069/0.105
DeepSeek-V4-pro	37	0.959 ± 0.05	0.060/0.104
GPT-5.4-mini	12	0.958 ± 0.07	0.058/0.099
GPT-5.4	12	0.875 ± 0.10	0.082/0.099
gpt-oss-120b	37	0.986 ± 0.03	0.053/0.104
gemma3-12b	37	0.986 ± 0.03	0.077/0.104

Table A4. Context ablation on HeartSteps ($K=35$, $n=37$): each row removes one prompt component from the full context. Removing prose components leaves classifier accuracy unchanged, while first-person framing collapses patient ranking and heterogeneity.

Prompt content	C2ST acc.	Spearman	heterogeneity
full context (baseline)	0.973	0.579 ± 0.12	0.919 ± 0.14
no domain description	0.973	0.606 ± 0.12	1.066 ± 0.17
no persona	0.973	0.555 ± 0.12	0.921 ± 0.14
no weather / location	0.973	0.520 ± 0.14	1.239 ± 0.20
no intervention text	0.973	0.605 ± 0.12	0.982 ± 0.14
structured fields only	0.973	0.597 ± 0.13	1.094 ± 0.18
history only	0.973	0.581 ± 0.13	1.321 ± 0.23
first-person framing	0.973	0.037 ± 0.14	0.032 ± 0.01

Table A5. Discrete-target reformulation on HeartSteps: the model predicts a behaviour class and the numeric outcome is reconstructed from the patient’s anchors. No target improves on the continuous control; coarse targets collapse onto the low-activity label (mean and upper tail destroyed) and the predicted class is essentially at chance.

Target	n	C2ST acc.	Spearman	mean sim/real	class acc.
continuous (control)	37	0.973 ± 0.05	0.579 ± 0.12	0.098/0.104	—
relative 5-bin	37	0.959 ± 0.05	0.624 ± 0.12	0.108/0.104	0.07
relative 3-bin	12	0.875 ± 0.11	0.755 ± 0.17	0.069/0.099	0.31
binary activity	12	0.958 ± 0.07	-0.096 ± 0.27	0.001/0.099	—
zero / positive	12	0.958 ± 0.07	-0.203 ± 0.35	0.004/0.099	—

Table A6. The emission layer on HeartSteps ($K=35$, $n=37$) against the prompt-only control. The four two-part variants are treated as tied and lower C2ST accuracy to ≈ 0.72 – 0.76 .

Configuration	C2ST acc.	Spearman	heterogeneity
prompt-only (control)	0.973 ± 0.05	0.579 ± 0.12	0.919 ± 0.14
two-part emission	0.743 ± 0.12	0.647 ± 0.11	0.835 ± 0.14
+ stateful variant	0.757 ± 0.09	0.654 ± 0.12	0.786 ± 0.11
+ action-noise hint	0.743 ± 0.08	0.644 ± 0.11	0.842 ± 0.14
+ stateful + hint	0.716 ± 0.09	0.611 ± 0.12	0.773 ± 0.11

Table A7. The no-LLM latent ablation on HeartSteps (Section 4.3): the identical emission layer driven by the LLM latent versus by the patient’s own anchor mean (zero model calls), as the anchor length K shrinks. The anchor-only baseline has lower C2ST at every K ; with little patient history, the LLM better preserves ranking, mean calibration and between-patient spread while remaining action-responsive. C2ST entries are pooled accuracy $\pm$ across-fold standard deviation.

K	C2ST $\downarrow$		Spearman		heterogeneity		mean	
	LLM	no-LLM	LLM	no-LLM	LLM	no-LLM	LLM	no-LLM
2	$0.784 \pm .039$	$0.581 \pm .051$	0.346	0.258	2.59	3.52	0.106	0.130
5	$0.608 \pm .126$	$0.554 \pm .107$	0.389	0.353	1.79	2.51	0.097	0.129
14	$0.676 \pm .135$	$0.500 \pm .178$	0.555	0.533	1.26	1.52	0.088	0.108
35	$0.743 \pm .122$	$0.541 \pm .111$	0.647	0.648	0.84	1.05	0.103	0.103

Table A8. Emission-parameter sensitivity on HeartSteps ($n=37$, $K=35$): C2ST accuracy, patient ranking, and heterogeneity around the fixed default. Accuracy responds mainly to the anchor weight w , not the dispersion ν . Spearman values are plug-in point estimates; a bootstrap over patients gives means ≈ 0.015 lower with identical ordering.

dispersion ν	anchor weight w	C2ST acc. $\downarrow$	Spearman	heterogeneity
0.7	0.35	0.932 ± 0.078	0.662	0.77
1.1	0.35	0.865 ± 0.108	0.651	0.61
0.9	0.50 (default)	0.743 ± 0.122	0.647	0.84
0.7	0.65	0.649 ± 0.073	0.559	1.11
1.1	0.65	0.622 ± 0.132	0.586	0.94

A.4 Benchmark-validity detail

Table A9. Benchmark validity ($\sigma=2$; 49 policies, three independent patient batches each): the overall correlation is moderate and significant, but agreement is good only within the lowest third and essentially zero within the highest third; the simulator systematically over-rates idle and over-messaging policies and under-rates the learned policies that are best on the reference target.

Policy subset	Spearman ρ (LLM vs. reference)	
all 49 (pooled)	0.66 [0.45, 0.81]	
per batch	0.59 ± 0.03	
lowest / middle third	0.54 / 0.11	
highest third	−0.03	

Policy family	Reference mean rank	LLM mean rank
online Q -learning (best family)	4	10
online + no-action blends	19	25
heuristic mixtures	23	19
do-nothing / random baselines	41	32

A.5 Statistical reporting protocol

Uncertainty is reported for the principal stochastic estimates and significance is tested where a defined reference exists. C2ST accuracy is computed under five-fold patient-disjoint cross-validation; we report pooled held-out accuracy with its across-fold standard deviation and test separability against the 0.5 chance level using a one-sided binomial test on the pooled held-out predictions. Marginal action effects and the simulator-versus-real comparisons are reported with bootstrap 95% confidence intervals over patients. Reinforcement-learning returns are reported as the mean $\pm$ standard deviation over training seeds and evaluation episodes (Table 3): evaluation uses eight seeds with eighty 50-step episodes per seed, while learned-policy variation covers nine offline fits (three corpora $\times$ three seeds) and three online training seeds. The benchmark-validity rank correlation is reported with a 95% bootstrap confidence interval (Table A9). For the realism results (Appendix Tables A2–A6) every configuration exceeds the 0.5 chance level at $p < 0.01$. The no-LLM controls in Table A7 are not significantly distinguishable at any anchor length; its LLM row at $K=5$ has $p = 0.040$. In Table A8, all configurations are distinguishable, but the final row has $p = 0.024$ rather than $p < 0.01$. Accuracy differences smaller than this cross-fold standard deviation (typically ± 0.05 – 0.12) are treated as ties rather than over-interpreted; this is why the factorial variants of the emission layer, which fall within one standard deviation of each other, are not claimed to differ (Section 4.1). The between-patient metrics (the Spearman patient-ranking correlation, the heterogeneity

ratio, and the autocorrelation error) are each reported with a bootstrap-over-patients standard deviation ($B=2000$ resamples of the 37 patients). These error bars quantify patient-sampling and cross-validation uncertainty, not variation across independent LLM generations.

A.6 Full list of limitations

Scope of the evidence. (1) The transfer and benchmark-validity results rest on a single environment with a known effect (StepCountJITAI); the data scarcity motivating this thesis meant that no real reinforcement-learning target with a ground-truth proximal effect was available, so the strongest RL claims are demonstrated on a synthetic target, with HeartSteps and HPTN 067 supplying the real-data evidence on the distributional and action-response side only. Step-CountJITAI is structurally simple, with two contexts, four actions, and hand-specified dynamics, so transfer on this controlled target may overstate performance in richer clinical environments. (2) The cross-domain test is likewise distributional: HPTN 067 shows the emission layer generalises, but no policy transfer is run on it, again for want of a known effect. The HeartSteps effect analysis is a marginal, patient-bootstrapped action contrast and does not condition on decision-time context.

Compute and cost. (3) The realism experiments vary the language model, but the reinforcement-learning and benchmark-validity results are not swept across models: re-running the full RL pipeline on the stronger commercial models was not feasible within the project’s inference budget because such models are costly at the volume the simulator requires. The benchmark-validity number is therefore model-specific. (4) Because the simulator issues one model call per decision point on a shared GPU cluster, the experiments were sized so as not to monopolise that shared resource: the RL results use nine offline fits and three online training seeds rather than the dozens that would tighten the intervals, and the benchmark-validity correlation, though estimated over 49 policies with 95% CI [0.45, 0.81], is still imprecise (a rank correlation over a few dozen policies). Both are reported with uncertainty for exactly this reason and neither is presented as a precise point estimate. Most realism configurations were generated once; broader repetition across generation seeds, decoding settings, model parameters and additional prompts was not feasible within the available time and inference budget.

Fidelity. (5) Even the best simulator remains distinguishable from real data. Anchoring and the two-part layer lower the classifier accuracy only to ≈ 0.72 , never to the 0.5 of indistinguishability, and the per-patient zero-inflated tail is not reproduced cell by cell. This imperfect realism did not prevent the simulator from training a policy that transfers (94% of the direct-training reference): a trajectory simulator can be a workable training environment well before it is a faithful one. (6) Calibration errors appear in different forms across the datasets: the simulator overstates HeartSteps’ near-zero action contrast and underestimates HPTN’s high outcome level (Section 4.3). (7) The realism and causal experiments anchor on each real patient’s history, whereas the RL experiments run a *self-anchored*

LLM world (Section 5); the two testbeds answer different questions and are not a single end-to-end pipeline. (8) The direct-training and oracle reference values are finite-run estimates rather than proven optima; each experimental configuration is evaluated under one principal prompt specification and decoding setup; and the public HeartSteps trajectories may fall within the model’s pre-training exposure, which we cannot rule out.

The benchmark’s resolution and observation mismatch. (9) On the paper-parameter-aligned target, benchmark validity has little resolution among the highest third ($\rho = -0.03$). (10) A plausible contributor is the observation mismatch: reference policies act on noisy inferred context, while LLM-world scoring supplies the current context directly; top-tier agreement improves as isolated context-inference noise is reduced (Section 4.4). (11) Per-policy LLM scores are also noisy (standard deviation about ± 0.35 across independent patient batches, comparable to the spread between policies), which is why a reliable estimate needs scoring over many patients and policies. The three batch-specific correlations are similar (0.59 ± 0.03), but broader replication would be needed to establish reproducibility.

Model capability, fine-tuning, and interpretability. (12) The model is used purely zero-shot, and the RL pipeline runs on a single open model. GPT-5.4 produced the best realism result in the matched twelve-patient comparison. The downstream transfer and benchmark experiments used Qwen3.5-122B, so whether that capability advantage extends to downstream results remains an empirical question. Fine-tuning the model on an aggregate of health-behaviour datasets, since no single one is large enough on its own, is a natural way to raise fidelity while preserving the zero-shot-at-deployment property, and is left to future work. (13) The simulator maintains an internal state and emits per-step reason codes, but we do not analyse these state or reason trajectories in depth; a systematic interpretation of why the simulator predicts as it does is left to future work.